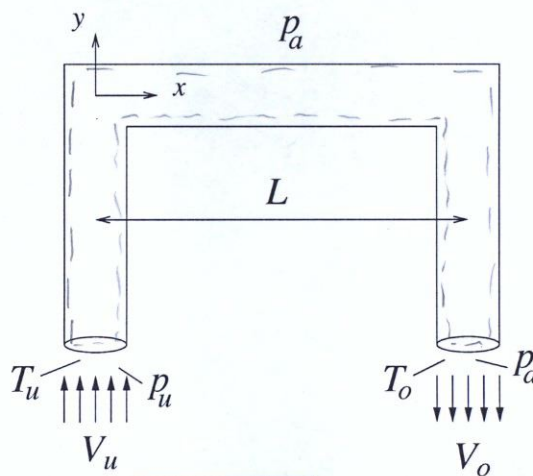


CL12 The gas circulating along the pipe of cross section A depicted in the figure discharges to the atmosphere, where the pressure is p_a . Measurements at a given upstream location give V_u , p_u , and T_u for the velocity, pressure and temperature, all uniform across the section. In the analysis that follows, it is assumed that the pipe is thermally insulated and that buoyancy forces have a negligible effect on the gas motion. The gas properties are defined by its gas constant R_g together with the ratio of specific heats $\gamma = c_p/c_v$.

- Obtain the values of the gas temperature and velocity at the outlet section T_o and V_o in terms of V_u , p_u , T_u , p_a , R_g and γ .
- Compute the two components of the force F_x and F_y acting on the stretch of pipe considered.
- Calculate the momentum on the pipe with respect to the origin of the coordinate system indicated in the figure.



$$\rho_u = \frac{p_u}{R_g T_u}$$

CONTINUITY

$$\rho_u V_u A = \rho_o V_o A \Rightarrow \frac{\rho_u V_u}{T_u} = \frac{\rho_o V_o}{T_o}$$

ENERGY

$$\int_{\xi_c} \rho \left(e + \frac{V^2}{2} \right) \bar{v} \cdot \bar{n} d\sigma = - \int_{\xi_c} p \bar{v} \cdot \bar{n} d\sigma + \int_{\xi_c} \bar{v} \cdot \bar{E}' \cdot \bar{n} d\sigma \Rightarrow \frac{\gamma}{\gamma-1} R_g T_u + \frac{1}{2} V_u^2 = \frac{\gamma}{\gamma-1} R_g T_o + \frac{1}{2} V_o^2$$

$$\bar{F} = - \int_{\xi_{int}} (p - p_a) \bar{n} d\sigma + \int_{\xi_{int}} \bar{E}' \cdot \bar{n} d\sigma$$



MOMENTUM CONSERVATION IN THE CONTROL VOLUME

$$\int_{\xi_c} \rho \bar{v} \cdot \bar{n} d\sigma = - \int_{\xi_c} (p - p_a) \bar{n} d\sigma + \int_{\xi_c} \bar{E}' \cdot \bar{n} d\sigma$$

$$-\rho_u V_u A V_u \bar{e}_y - \rho_o V_o A V_o \bar{e}_y = - \int_{\xi_{ur}} (p - p_a) \bar{n} d\sigma + (p_u - p_a) \bar{e}_y A + \int_{\xi_{ur}} \bar{E}' \cdot \bar{n} d\sigma$$

$$\bar{F} = [\rho_u V_u A (V_u + V_o) + (p_u - p_a) A] \bar{e}_y$$

$$\bar{M} = - \int_{\xi_{int}} (p - p_a) \bar{x} \wedge \bar{n} d\sigma + \int_{\xi_{int}} \bar{x} \wedge \bar{E}' \cdot \bar{n} d\sigma$$

CONSERVATION OF ANGULAR MOMENTUM IN THE CONTROL VOLUME

$$\int_{\xi_c} (\bar{x} \wedge \rho \bar{v}) \cdot \bar{n} d\sigma = - \int_{\xi_c} \bar{x} \wedge p \bar{n} d\sigma + \int_{\xi_c} \bar{x} \wedge \bar{E}' \cdot \bar{n} d\sigma$$

$$-\rho_u V_u A L V_o \bar{e}_z = - \int_{\xi_{ur}} \bar{x} \wedge (p - p_a) \bar{n} d\sigma + \int_{\xi_{ur}} \bar{x} \wedge \bar{E}' \cdot \bar{n} d\sigma$$

$$\bar{M} = \rho_u V_u A L V_o \bar{e}_z$$